

Calculus II

Area swept by cardioid $r = a + b \sin(\theta)$,
 $r = a + b \cos(n\theta)$

Todor Milev

2019

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

$$\begin{aligned}
 A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta
 \end{aligned}$$